Assignment 6 – Binary Files Indexing

Spring 2026

Due: Saturday, May 2 11:59 pm

The goal is to implement indexing on a binary file containing gene sample records. You will use the provided input file, samples.bin, which includes **500,000 gene sample records**. Each record adheres to a predefined structure and includes a species code.

Instructions:

1. **Read and sort** the records in samples.bin by **species code**.
2. Save the sorted data in a new binary file named sorted_samples.bin.
3. Create and apply an **indexing scheme** based on the species code to enable efficient retrieval.

Species Codes:
The records use one of the following 5 unique species codes:

- 0 – *H_SAP* (Homo sapiens)
- 1 – *M_MUS* (Mus musculus)
- 2 – *D_MEL* (Drosophila melanogaster)
- 3 – *E_COL* (Escherichia coli)
- 4 – *A_THA* (Arabidopsis thaliana)

To-do

Download the base code **geneRecords** from D2L and complete the following tasks:

Task 1: Complete the implementation in **main()** and for **searchSample()**

Task 2: Complete the implementation for **your chosen sorting algorithm.**

Task 3: Complete the **p_index()** function.

Task 4: Complete the **p_displayResearcher()** function.

Task 5: Complete the **p_updateResearcher()** function.

Task 6: Complete the **p_deleteSample()** function.

Task 7: Complete the **p_printRange()** function.

Task 8: Use make files to automate compiling and running your program (use version 5)

Total Points (100)

- Code runs and works as expected – Task 1: 15 points
- Code runs and works as expected – Task 2: 20 points
- Code runs and works as expected – Task 3: 15 points
- Code runs and works as expected – Task 4: 10 points
- Code runs and works as expected – Task 5: 10 points
- Code runs and works as expected – Task 6: 10 points
- Code runs and works as expected – Task 7: 10 points
- Used make files – Task 8: 5 points
- Available on GitHub: 5 points

Deliverables:

- A zipped folder named A6 that contains all the files
  - C++ files and header files
  - README.txt file
  - A screenshot showing the functions work.
- Published to D2L dropbox and GitHub under the folder where you added me as a collaborator.